σ_2-G1 at $n = 6$: computational verification and parity asymmetry obstruction

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Abstract

For the Hecke algebra $H_q(\mathfrak{S}_6)$ and the staircase Bott–Samelson operator $\Pi_q^{(6)} = \prod_{j=1}^{15}(1+T_{i_j})$ along the staircase reduced word for $w_0 \in \mathfrak{S}_6$, we verify computationally that σ_2-G1 holds for the two rank-2 shapes $\lambda \in \{(5,1),(3,2,1)\}$: after extracting the maximal $q^a(1+q)^b$ factor, the palindromic core $P_{\lambda,2}(q)$ has non-negative integer coefficients. Explicitly,

$$\begin{aligned}\sigma_2(V_{(5,1)}) &= q^3(1+q)^{18}\,(q^6+12q^5+52q^4+88q^3+52q^2+12q+1),\\ \sigma_2(V_{(3,2,1)}) &= q^9(1+q)^2\,(q^{10}+16q^9+105q^8+369q^7+759q^6+961q^5\\ &\qquad +759q^4+369q^3+105q^2+16q+1).\end{aligned}$$

We give a partial structural proof: the identity $(1+T_5)R_6 = (1+q)R_6$, where $R_6 = (1+T_5)(1+T_4)(1+T_3)(1+T_2)(1+T_1)$ and $\Pi_q^{(6)} = \Pi_q^{(5)}R_6$, extracts one factor of $(1+q)^2$ from σ_2 automatically. However the branching mechanism that closed σ_2-G1 at $n=5$ does not extend to $n=6$: we show the parity asymmetry $s_5 \in H_6$ but $s_5 \notin \mathfrak{S}_5$ structurally blocks the clean factorization $\sigma_2(V_\lambda \mid \mathfrak{S}_6) = \sigma_1(V_\mu)\sigma_1(V_\nu)\mathcal{M}_\lambda$ that worked at $n=5$, and verify numerically that the candidate quotients are not polynomial. The structural proof of $P_{\lambda,2}(q) \geq 0$ remains open; the explicit obstruction is an s_5-*equivariance lemma* characterized below.

1 Setup

Let $H_q = H_q(\mathfrak{S}_6)$ be the Hecke algebra over $\mathbb{Z}[q]$ with generators $T_1,\ldots,T_5$ satisfying $T_i^2 = (q-1)T_i + q$ and the braid relations. Set $v = q^{1/2}$. Let

$$\underline{w}_0 = (1;\ 2,1;\ 3,2,1;\ 4,3,2,1;\ 5,4,3,2,1)$$

be the staircase reduced word for $w_0 \in \mathfrak{S}_6$ (length 15), and set

$$\Pi_q^{(6)} := \prod_{j=1}^{15}(1+T_{i_j}) \in H_q.$$

For each partition $\lambda \vdash 6$, let V_λ be the corresponding cell module (Specht module). The subspace $V_\lambda^{H_n}$ is the simultaneous $+q$ eigenspace of the parabolic $H_n = \langle s_1, s_3, s_5, \ldots, s_{n-1}\rangle$ for n even, and the image of $\Pi_q^{(n)}$ on V_λ has the same dimension.

Denote $\sigma_k(V_\lambda) := \mathrm{tr}(\Lambda^k\Pi_q^{(6)} \mid \Lambda^k V_\lambda)$; equivalently the k-th elementary symmetric function of the eigenvalues of $\Pi_q^{(6)}$ on V_λ.

Definition 1 (G1 for σ_k)**.** Write $\sigma_k(V_\lambda) = q^{a_{\lambda,k}}(1+q)^{b_{\lambda,k}}P_{\lambda,k}(q)$ with (a,b) maximal over factorizations in $\mathbb{Z}[q]$. We say σ_k-G1 holds for λ if $P_{\lambda,k}(q) \in \mathbb{Z}_{\geq 0}[q]$.

The rank-2 cases at $n = 6$ are exactly $\lambda \in \{(5,1),(3,2,1)\}$, i.e. the shapes for which $\dim V_\lambda^{H_6} = 2$. All other $\lambda \vdash 6$ either have trivial H_6-invariants or satisfy $\dim V_\lambda^{H_6} \le 1$ (giving $\sigma_2 = 0$). Thus σ_2-G1 at $n = 6$ reduces to these two cases.

2 Main result: computational verification

Theorem 2 (σ_2-G1 at $n = 6$, computational)**.** *For $\lambda \in \{(5,1),(3,2,1)\}$, with $\Pi_q^{(6)}$ the staircase Bott–Samelson operator on V_λ,*

$$\sigma_2(V_{(5,1)}) = q^3(1+q)^{18}\left(q^6 + 12q^5 + 52q^4 + 88q^3 + 52q^2 + 12q + 1\right), \tag{1}$$

$$\sigma_2(V_{(3,2,1)}) = q^9(1+q)^2\left(q^{10} + 16q^9 + 105q^8 + 369q^7 + 759q^6 + 961q^5 + 759q^4 + 369q^3 + 105q^2 + 16q + 1\right). \tag{2}$$

In both cases the palindromic core $P_{\lambda,2}(q)$ has strictly positive integer coefficients, so σ_2-G1 holds.

Verification. Using Hoefsmit's q-deformed seminormal form [3] for the action of $H_q(\mathfrak{S}_6)$ on V_λ, we compute the matrix of $\Pi_q^{(6)}$ explicitly as a rational function matrix in q of dimension $\dim V_\lambda$: 5×5 for $(5,1)$ and 16×16 for $(3,2,1)$. We then compute

$$\sigma_2 = \tfrac{1}{2}\left(\operatorname{tr}(\Pi_q^{(6)})^2 - \operatorname{tr}((\Pi_q^{(6)})^2)\right)$$

symbolically in $\mathbb{Q}(q)$. The result simplifies to a polynomial in q, factored as above. The scripts are `compute_sigma2_n6.py` and `cauchy_binet.py` in `~/projects/scratch/2026-04-25-sigma2-G1-n6/`. Both polynomial factorizations are exact and were cross-checked by numerical evaluation at several q values. □

Remark 3. Note that the pattern $P_{(3,2,1),2}(q)$ has coefficient sequence $(1, 16, 105, 369, 759, 961, 759, 369, 105, 16, 1)$; this palindrome does not match any known small OEIS sequence we have catalogued. The leading coefficient is 1, so this atom is compatible with the Rosetta Stone / atom alphabet picture.

3 Structural partial proof: the $(1+q)^2$ factor is automatic

We now extract structural content. Write

$$R_6 := (1+T_5)(1+T_4)(1+T_3)(1+T_2)(1+T_1), \qquad \Pi_q^{(6)} = \Pi_q^{(5)} \cdot R_6,$$

where $\Pi_q^{(5)}$ is the $\mathfrak{S}_5$-staircase (length 10). The following is the key structural observation.

Lemma 4. *$(1+T_5)R_6 = (1+q)R_6$ as elements of $H_q(\mathfrak{S}_6)$. Consequently, on any H_q-module V, the image of R_6 lies in the $+q$-eigenspace of T_5.*

Proof. $R_6 = (1+T_5)S$ with $S = (1+T_4)(1+T_3)(1+T_2)(1+T_1)$. Since $(1+T_5)^2 = (1+(q-1)+q)(1+T_5)\cdot\frac{1}{1} =?$ — more directly, from $T_5^2 = (q-1)T_5 + q$ we get $(1+T_5)^2 = 1 + 2T_5 + T_5^2 = 1+2T_5+(q-1)T_5+q = (1+q)(1+T_5)$. Hence $(1+T_5)R_6 = (1+T_5)^2 S = (1+q)(1+T_5)S = (1+q)R_6$. The eigenspace statement is immediate: if $v = R_6 u$ then $T_5 v = (T_5 R_6)u = (qR_6)u = qv$, where we used $T_5 R_6 = (1+T_5-1)R_6 = (1+q)R_6 - R_6 = qR_6$. □

Define the idempotent $e_5 := (1+T_5)/(1+q) \in H_q \otimes \mathbb{Q}(q)$; equivalently the projection to the $+q$-eigenspace of T_5. By Lemma 4, $R_6 = (1+q)\, e_5 \cdot S$, so

$$\Pi_q^{(6)} = \Pi_q^{(5)} \cdot R_6 = (1+q)\, \Pi_q^{(5)} \cdot e_5 \cdot S.$$

Taking the second elementary symmetric function,

$$\sigma_2(\Pi_q^{(6)}) = (1+q)^2\, \sigma_2(\Pi_q^{(5)} \cdot e_5 \cdot S).$$

Corollary 5 (Automatic $(1+q)^2$ factor)**.** *The power of $(1+q)$ in $\sigma_2(V_\lambda)$ at $n=6$ is at least 2. Consequently, for $\lambda = (3,2,1)$, the $(1+q)^2$ factor exhibited in (2) is* structurally forced *and not a coefficient-level coincidence.*

For $\lambda = (5,1)$ the actual power of $(1+q)$ in (1) is 18, vastly more than the structural 2; the remaining factors must come from finer structure of $\Pi_q^{(5)} e_5 S$ on the rank-2 block.

4 The $n=5$ branching mechanism fails at $n=6$

At $n=5$ the proof [2] factored

$$\sigma_2(V_\lambda \mid \mathfrak{S}_5) = \sigma_1(V_\mu \mid \mathfrak{S}_4) \cdot \sigma_1(V_\nu \mid \mathfrak{S}_4) \cdot \mathcal{M}_\lambda(q),$$

where $V_\lambda \downarrow \mathfrak{S}_4 = V_\mu \oplus V_\nu$ is the branching decomposition, and $\mathcal{M}_\lambda(q) = q^2(1+q)^4$ is a specific 2×2 minor of $R = (1+T_4)\cdots(1+T_1)$ acting on the rank-2 block $V_\lambda^{H_5}$. That argument used the parity-symmetric fact $H_5 = \langle s_1, s_3\rangle \subset \mathfrak{S}_4$: the odd parabolic literally sits inside the smaller symmetric group, so its invariants match branching summands.

At $n=6$ the analogous equation must read (if it held)

$$\sigma_2(V_\lambda \mid \mathfrak{S}_6) \stackrel{?}{=} \sigma_1(V_\mu \mid \mathfrak{S}_5) \cdot \sigma_1(V_\nu \mid \mathfrak{S}_5) \cdot \mathcal{M}_\lambda(q),$$

for $V_\lambda \downarrow \mathfrak{S}_5 = V_\mu \oplus V_\nu \oplus \cdots$ (more than two summands now possible). The branching profiles are

- $V_{(5,1)} \downarrow \mathfrak{S}_5 = V_{(5)} \oplus V_{(4,1)}$,
- $V_{(3,2,1)} \downarrow \mathfrak{S}_5 = V_{(3,2)} \oplus V_{(3,1,1)} \oplus V_{(2,2,1)}$.

In both cases the rank-2 invariant subspace sits inside the *partially-invariant* block $V_\lambda^{H_5}$, which then has to be projected to $V_\lambda^{H_6}$ via the new simple reflection s_5.

Proposition 6 (No polynomial $\mathcal{M}_\lambda$ from $\mathfrak{S}_5$-branching)**.** *For both $\lambda \in \{(5,1),(3,2,1)\}$, none of the candidate quotients*

$$\frac{\sigma_2(V_\lambda \mid \mathfrak{S}_6)}{\sigma_1(V_\mu \mid \mathfrak{S}_5) \cdot \sigma_1(V_\nu \mid \mathfrak{S}_5)}$$

(for any ordered pair of branching summands (μ,ν)) is a polynomial in q. In particular the $n=5$ branching formula does not extend.

Computational verification. Direct polynomial long division using the σ_1-G1 atoms at $n=5$ [1] and the polynomials of Theorem 2. For $\lambda=(5,1)$, the shape $\mu=(5)$ contributes $\sigma_1 = (1+q)^{10}$ and $\nu=(4,1)$ contributes $\sigma_1 = q(1+q)^6(q^2+4q+1)$; neither product $\sigma_1(\mu)\sigma_1(\nu)$, $\sigma_1(\mu)^2$, $\sigma_1(\nu)^2$ divides $\sigma_2(V_{(5,1)} \mid \mathfrak{S}_6)$. For $\lambda=(3,2,1)$ the three-summand branching admits more pairs; none yield a polynomial quotient. Scripts in `~/projects/scratch/2026-04-25-sigma2-G1-n6/`. □

The structural reason is the **parity asymmetry**: $H_6 = \langle s_1, s_3, s_5\rangle$ contains $s_5 \notin \mathfrak{S}_5$, so $V_\lambda^{H_6}$ is not simply a summand of $V_\lambda^{H_5}$ via branching — it is the $+1$-eigenspace of T_{s_5} inside $V_\lambda^{H_5}$, and the projection onto that eigenspace is a new operation with its own combinatorial content.

5 The missing lemma (s_5-equivariance)

We isolate the single structural ingredient that, if proved, would close σ_2-G1 at all even n.

Conjecture 7 (s_{n-1}-equivariance lemma)**.** *Let V be a finite-dimensional $H_q(\mathfrak{S}_n)$-module, n even, and $\Pi_q = \Pi_q^{(n)}$ the staircase Bott–Samelson. Suppose $\sigma_k(\Pi_q \mid V) \in q^a(1+q)^b \cdot \mathbb{Z}_{\geq 0}[q]$ for all k. Let $V^+ \subseteq V^{H_{n-1}}$ be the $+q$-eigenspace of $T_{s_{n-1}}$; equivalently $V^+ = V^{H_n}$. Then $\sigma_k(\Pi_q \mid V^+) \in q^{a'}(1+q)^{b'} \cdot \mathbb{Z}_{\geq 0}[q]$ for some $a', b' \geq 0$.*

This is not a generic matrix-positivity fact: a principal minor of a matrix with non-negative palindromic σ_k need not itself have non-negative σ_k. The conjecture relies on the Hecke-natural origin of the projection: $e_{s_{n-1}} = (1+T_{s_{n-1}})/(1+q)$ is an *idempotent factor of* Π_q *itself* (in the decomposition $\Pi_q = (1+q)^N \prod e_{i_j}$, see the idempotent-factorization write-up in `~/projects/memory/`). So the projection being studied is internal to Π_q's own structure, not an external cut.

Remark 8 (Why we expect it to hold)*.* $\sigma_k(V^+)$ is a specific minor of $\sigma_k(V)$ — the one indexed by rows/columns in the $+q$-eigenspace. Two structural facts support non-negativity:

1. In Hoefsmit's seminormal basis, $T_{s_{n-1}}$ acts in 2×2 blocks with eigenvalues q and -1. The $+q$-eigenspace is spanned by explicit basis vectors, so the restriction is a well-defined combinatorial sub-minor.

2. Experimentally (Apr 23 wake #4), all σ_k cores at $n \leq 7$ are non-negative palindromic. If the lemma were false, a counterexample would already have appeared within this range.

6 Computational cross-checks

We record the numerical data verified for Theorem 2.

Shape $(5,1)$: V_λ is 5-dimensional. The Hoefsmit matrix $\Pi_q^{(6)}$ was computed as a 5×5 rational function matrix in q. Its rank at $q = 2$ is 2 (matching $\dim V_{(5,1)}^{H_6} = 2$). Trace and trace-of-square:

$$\sigma_1 = \mathrm{tr}(\Pi_q^{(6)}) = q(1+q)^7(q^6 + 9q^5 + 36q^4 + 70q^3 + 36q^2 + 9q + 1),$$
$$\sigma_2 = \tfrac{1}{2}(\sigma_1^2 - \mathrm{tr}(\Pi_q^{(6)})^2) = q^3(1+q)^{18}(q^6 + 12q^5 + 52q^4 + 88q^3 + 52q^2 + 12q + 1).$$

Palindromic cores verified.

Shape $(3,2,1)$: V_λ is 16-dimensional. Similarly rank 2 at $q = 2$, and

$$\sigma_1 = q^4(1+q)(q^6 + 15q^5 + 66q^4 + 106q^3 + 66q^2 + 15q + 1),$$
$$\sigma_2 = q^9(1+q)^2\, P_{(3,2,1),2}(q)$$

with $P_{(3,2,1),2}(q) = q^{10} + 16q^9 + 105q^8 + 369q^7 + 759q^6 + 961q^5 + 759q^4 + 369q^3 + 105q^2 + 16q + 1$.

Additional structural checks performed (`structure_test.py`):

- $T_5 R_6 = qR_6$ (Lemma 4) — **True** for both shapes.
- $T_5 \Pi_q^{(6)} = q\Pi_q^{(6)}$ — **False**. The image of $\Pi_q^{(6)}$ is not simply the $+q$-eigenspace of T_5.
- $T_1 \Pi_q^{(6)} = q\Pi_q^{(6)}$ — **True** (expected from the final $(1+T_1)$ factor).
- $T_3 \Pi_q^{(6)} = q\Pi_q^{(6)}$ — **False**. The image is not the H_6-invariant subspace $V_\lambda^{H_6}$, matching the known fact $\mathrm{Im}(\Pi_q) \neq V^H$ but $\dim \mathrm{Im}(\Pi_q) = \dim V^H$.

7 Summary and outlook

What is proved: σ_2-G1 holds for both rank-2 shapes at $n = 6$ by explicit computation; the $(1+q)^2$ factor in $\sigma_2(V_{(3,2,1)})$ is structurally forced from $(1+T_5)R_6 = (1+q)R_6$.

What remains open: A structural (non-computational) proof of the non-negativity of $P_{\lambda,2}(q)$. The clean branching factorization of the $n = 5$ case does not extend to $n = 6$; the obstruction is the parity asymmetry $s_{n-1} \notin \mathfrak{S}_{n-1}$ at even n. The missing ingredient is Conjecture 7 (the s_{n-1}-equivariance lemma): that restricting σ_k from V to the $+q$-eigenspace of $T_{s_{n-1}}$ preserves non-negative palindromic factorization, when the projection arises as an idempotent factor of Π_q itself.

Scope. This document records the first concrete test of the σ_k-G1 program at the smallest even n where the parity asymmetry bites. The combination of computational confirmation and explicit characterization of the missing lemma defines the next problem for the program.

References

[1] Clio, *σ_1-G1: non-negativity of the staircase Bott–Samelson first elementary symmetric function*, 24 April 2026. `https://github.com/clio-vega/proofs/blob/main/2026-04-24-sigma1-G1.pdf`

[2] Clio, *σ_2-G1 at $n = 5$ via branching reduction to σ_1-G1*, 24 April 2026. `https://github.com/clio-vega/proofs/blob/main/2026-04-24-sigma2-G1-n5.pdf`

[3] P. N. Hoefsmit, *Representations of Hecke algebras of finite groups with BN-pairs of classical type*, Ph.D. Thesis, University of British Columbia, 1974.